

SEMINAR III • WEEK 7 • FOUNDATIONS

Letters That Stand for Numbers

A variable isn't a mystery to solve every time you see one — sometimes it's just a value that changes, like hours worked or number of items bought.

SESSION A

Naming the Parts, and Evaluating

WHAT'S IN AN EXPRESSION?

The diagram shows the expression $3x + 7$. Below the number 3 is a vertical line pointing to the label "coefficient". Below the letter x is a vertical line pointing to the label "variable". Below the number 7 is a vertical line pointing to the label "constant".

The **coefficient** multiplies the variable. The **constant** stands alone. An expression has no equal sign — an equation does.

A TRAP WORTH KNOWING

Does a variable always mean "find this missing number"?

No — a variable can represent a quantity that changes, like hours worked. Not every expression needs to be solved.

TRANSLATING WORDS TO EXPRESSIONS

"Five less than a number" $\rightarrow x-5$, **not** $5-x$. "Less than" reverses the order you might expect.

Try it with real situations — a car rental with a per-day fee, a phone plan with a base cost plus usage.

A TRAP WORTH KNOWING

Can $3x + 7$ be "solved"?

No equal sign, nothing to solve for. It can be simplified, or evaluated at a given x — not solved.

SUBSTITUTION

Evaluate $5x - 3$ when $x = 4$: $5(4) - 3 = 20 - 3 = 17$.

Replace *every* occurrence of the variable — in $3x + 2x$, both x 's get replaced. Order of operations still applies after you substitute.

BEFORE WE MOVE ON

1. Identify the coefficient, variable, and constant
2. Translate a phrase into an expression
3. Substitute and evaluate with a negative value

SESSION B

Combining Like Terms, and Distributing

LIKE TERMS

Same variable, same power — only then can they combine.

$4x + 3x^2$ is **not** $7x$ or $7x^2$ — different powers, not like terms.

THE DISTRIBUTIVE PROPERTY

$$3(x + 4) = 3x + 12$$

The outside factor multiplies **every** term inside.

$3(x+4) = 3x+4$ is wrong — only the first term got multiplied. And a negative factor distributes too: $-2(x+6) = -2x-12$, not $-2x+6$.

PUTTING IT TOGETHER

Simplify $3(x + 2) + 4x$: distribute first $\rightarrow$
 $3x+6+4x$. Combine like terms $\rightarrow$ **$7x + 6$** .

BEFORE WE MOVE ON

Distribute and simplify one expression with a negative factor.

Then: mixed practice, error analysis, Week 7 check, reflection.

TRANSITION TO WEEK 8

Equations

What does the unknown represent, and what operation will undo what's happening to it?